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Battery pack and electric vehicle

Abstract

Provided is a battery pack including a pair of output terminals, a control unit, and a plurality of battery units, where each of the plurality of battery units includes a plurality of battery blocks connected in series, a positive electrode terminal, and a negative electrode terminal, the battery block includes one battery or a plurality of batteries connected in parallel, a fuse is connected between at least one output terminal of the pair of output terminals and a positive electrode terminal or a negative electrode terminal corresponding to the one output terminal, and the fuse is allowed to be fused by the control unit.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a battery pack and an electric vehicle.

BACKGROUND ART

(2) In recent years, secondary batteries have been expanding in application. For example, lithium ion secondary batteries, which are typical examples of secondary batteries, have been expanding in application not only to various electronic devices but also to automobiles, motorcycles, electric flight vehicles, and the like. Depending on the application, not one (single cell) but a plurality of

lithium ion secondary batteries may be used. When a plurality of lithium ion secondary batteries are used, an abnormality such as an internal short circuit may be caused in a predetermined battery of the lithium ion secondary batteries. Patent Document 1 below describes a battery module in which a lithium ion secondary battery with such an abnormality caused is separated in a circuit sense with the use of a fuse so as to secure safety.
PRIOR ART DOCUMENT
Patent Document
(3) Patent Document 1: International Publication No. WO 2012/014350
SUMMARY OF THE INVENTION
Problem to be Solved by the Invention
(4) The technique described in Patent Document 1 fails, however, to control the timing at which the fuse is fused, and thus has the possibility of failing to separate, in a circuit sense, the lithium ion secondary battery with an abnormality caused.
(5) Accordingly, an object of the present invention is to provide a battery pack and an electric vehicle that allow a fuse to be fused at an appropriate timing to allow a secondary battery with an abnormality caused to be reliably separated in a circuit sense.
Means for Solving the Problem
(6) The present invention provides a battery pack including a pair of output terminals, a control unit, and a plurality of battery units, where each of the plurality of battery units includes a plurality of battery blocks connected in series, a positive electrode terminal, and a negative electrode terminal, the battery block includes one battery or a plurality of batteries connected in parallel, a fuse is connected between at least one output terminal of the pair of output terminals and a positive electrode terminal or a negative electrode terminal corresponding to the one output terminal, and the fuse is allowed to be fused by the control unit.
Advantageous Effect of the Invention
(7) According to at least an embodiment of the present invention, the fuse can be fused at an appropriate timing, thus allowing a secondary battery with an abnormality caused to be reliably separated in a circuit sense. It is to be noted that the contents of the present invention are not to be construed as being limited by the effects illustrated in this specification.

Description

BRIEF EXPLANATION OF DRAWINGS
(1) FIG. **1** is a diagram referred to in the description of problems to be considered in an embodiment.
(2) FIG. **2** is a diagram referred to in the description of problems to be considered in an embodiment.
(3) FIG. **3** is a diagram referred to in the description of problems to be considered in an embodiment.
(4) FIG. **4** is a diagram referred to in the description of problems to be considered in an embodiment.
(5) FIG. **5** is a diagram illustrating a schematic configuration example of a battery pack according to an embodiment.
(6) FIGS. **6**A and **6**B are diagrams illustrating a configuration example of a heater fuse according to an embodiment.
(7) FIG. **7** is a diagram illustrating a detailed configuration example of a battery pack according to an embodiment.
(8) FIG. **8** is a diagram for explaining an application example.
MODE FOR CARRYING OUT THE INVENTION

(9) Hereinafter, embodiments and the like of the present invention will be described with reference to the drawings. It is to be noted that the description will be provided in the following order. <Problems to be Considered in Embodiment> <Embodiment> <Modification Example> <Application Example>
(10) The embodiment and the like described below are preferred specific examples of the present invention, and the contents of the present invention are not to be considered limited to the embodiments and the like.
Problems to be Considered in Embodiment
(11) First, for facilitating understanding of the present invention, problems to be considered in an embodiment will be described.
(12) FIG. **1** is a diagram illustrating a configuration example of a common battery pack (battery pack **1**). It is to be noted that the arrows illustrated in FIG. **1** schematically illustrate the flows of currents, and the thicknesses thereof indicate the magnitudes of the currents. The thicker arrow indicates a larger current, whereas the thinner arrow indicates a smaller current.
(13) The battery pack **1** includes a battery section **2**, a positive electrode terminal TA led out from the positive electrode side of the battery section **2**, and a negative electrode terminal TB led out from the negative electrode side of the battery section **2**.
(14) The battery section **2** includes, for example, eight secondary batteries and seven fuses. The secondary battery is, for example, a single cell (hereinafter, simply referred to as a battery.) of a lithium ion secondary battery. The battery **11** and the battery **12** are connected in series, the battery **13** and the battery **14** are connected in series, the battery **15** and the battery **16** are connected in series, and battery **17** and battery **18** are connected in series.
(15) The connection midpoint between the battery **11** and the battery **12** and the connection midpoint between the battery **13** and the battery **14** are connected via the fuse **21**. In addition, the connection midpoint between the battery **13** and the battery **14** and the connection midpoint between the battery **15** and the battery **16** are connected via the fuse **22**. In addition, the connection midpoint between the battery **15** and the battery **16** and the connection midpoint between the battery **17** and the battery **18** are connected via the fuse **23**.
(16) The fuse **25** is connected between the positive electrode terminal TA and the positive electrode side of the battery **11**. The fuse **26** is connected between the positive electrode terminal TA and the positive electrode side of the battery **13**. The fuse **27** is connected between the positive electrode terminal TA and the positive electrode side of the battery **15**. The fuse **28** is connected between the positive electrode terminal TA and the positive electrode side of the battery **17**.
(17) In this regard, an example in which the battery **18** is internally short-circuited is considered. The internal short-circuit causes a short-circuit current to flow. Specifically, currents discharged from the battery **12**, the battery **14**, and the battery **16** flows through a loop RP**1** on the lower stage side of the battery section **2** as illustrated in FIG. **1**. The current C**12** discharged from the battery **12** passes through the fuse **21**, and then, is combined with the current C**14** discharged from the battery **14** to become a current CA. The current CA passes through the fuse **22**, and then, is combined with the current C**16** discharged from the battery **16** to become a current CB. The current CB flows through the fuse **23**.
(18) The current flowing through the fuse **23** is larger than the current flowing through each of the fuse **21** and the fuse **22** (for simplification, the current flowing through the fuse **23** is three times as large as the current flowing through the fuse **21**). Thus, as illustrated in FIG. **2**, the fuse **23** is fused prior to the fuse **21** and the fuse **22**.
(19) Thereafter, as illustrated in FIG. **2**, the currents discharged from the batteries **11** to **16** flow through a loop RP**2** of the battery section **2**. The currents (current CC) flowing through the fuse **25**, the fuse **26**, and the fuse **27** are substantially equal in magnitude. In contrast, the current flowing through the fuse **28** is a current **3**CC obtained by combining the currents flowing through each of the fuse **25**, the fuse **26**, and the fuse **27**. Thus, as illustrated in FIG. **3**, the fuse **28** is fused prior to

each of the fuse **25**, the fuse **26**, and the fuse **27**. As a result, the battery **18** internally short-circuited is separated as an open circuit, that is, in a circuit sense in the battery pack **1**. Accordingly, the battery **18** is not allowed to be charged or discharged, thus allowing the battery **18** to be prevented from generating heat or being ignited, and allowing the safety of the battery pack **1** to be secured.
(20) It is to be noted that if the battery other than the battery **18** is internally short-circuited, the safety of the battery pack **1** is similarly secured.
(21) FIG. **4** is a diagram illustrating a configuration example of a battery pack (battery pack **1**A) according to another example of a common battery pack. The battery pack **1**A differs from the battery pack **1** in terms of configuration in that the battery pack **1**A has a battery section **2**A instead of the battery section **2**. The battery section **2**A includes the above-described batteries **15** to **18**, and fuse **23**, fuse **27**, and fuse **28** (each configuration has the same connection relationship as that in the battery section **2**).
(22) An example in which the battery **18** is internally short-circuited is considered. As illustrated in FIG. **4**, after fuse **23** is fused by a current C**16** discharged from the battery **16**, a current CC discharged from the battery **15** and the battery **16** also flows through the fuse **28** via the fuse **27**.
(23) The currents flowing through the fuse **27** and the fuse **28** are equal in magnitude. For this reason, which of the fuse **27** and the fuse **28** is fused first is not known. For example, if the fuse **27** is fused prior to the fuse **28**, the battery **18** internally short-circuited fails to be separated in a circuit sense, thus causing a problem that the battery **18** continues to be charged and discharged. It is not possible to predict which battery is internally short-circuited, and it is thus practically impossible to make only a specific fuse into a fuse that has a characteristic of being likely to be fused. Accordingly, it is desirable to be capable of controlling the timing of fusing the fuse. In addition, for example, as in the circuit illustrated in FIG. **4**, when the number of series-connected battery groups connected in parallel is two, it is desirable to be capable of controlling the timing of fusing the fuse connected to each of the battery groups. An embodiment of the present invention will be described in view of the foregoing respects.
Embodiment
Definitions of Terms Used in this Specification
(24) First, the definitions of terms used in this specification will be described.
(25) “Battery” . . . means a so-called single battery (single cell). It is to be noted that a single cell of a lithium ion secondary battery will be described as an example of the battery in the present embodiment, other batteries may be employed.
(26) “Battery block” . . . means a block including one battery or a plurality of batteries connected in parallel.
(27) “Battery unit” . . . means a unit that has a plurality of battery blocks connected in series. Each of different battery units has the same number of battery blocks.
(28) “Battery section” . . . means a configuration including a plurality of battery units.
(29) [Battery Pack According to Present Embodiment]
(30) (Schematic Configuration)
(31) FIG. **5** is a diagram illustrating a schematic configuration example of a battery pack (battery pack **10**) according to the present embodiment. The battery pack **10** includes a battery section **20**. A positive electrode terminal TA is led out from the positive electrode side of the battery section **20**. A negative electrode terminal TB is led out from the negative electrode side of the battery section **20**. The positive electrode terminal TA and the negative electrode terminal TB correspond to a pair of output terminals.
(32) The battery pack **10** includes a battery unit **30**, a control integrated circuit (IC) **35**, a heater fuse **38**, a battery unit **40**, a control IC **45**, a heater fuse **48**, and a fuse **50**. The configuration including the battery unit **30** and the heater fuse **38** and the configuration including the battery unit **40** and the heater fuse **48** are connected in parallel. It is to be noted that according to the present embodiment, each of the control IC **35** and the control IC **45** corresponds to a control unit.

(33) The battery unit **30**, which is an example of the first battery unit, has the form of a battery block **31** and a battery block **32** connected in series. Specific configuration examples of the battery block **31** and battery block **32** will be described later. In addition, the battery unit **30** has a positive electrode terminal T**1** and a negative electrode terminal T**2**.
(34) The control IC **35** is connected to the battery unit **30** and the heater fuse **38**. The control IC **35** determines the state of each of the battery block **31** and battery block **32**. For example, the control IC **35** measures the voltage for each of the battery block **31** and battery block **32**, and determines whether each battery block is overcharged or not. Then, if there is any battery block overcharged, the control IC **35** controls the current to the heater of the heater fuse connected in series to the battery block, that is, the heater fuse **38**. This control allows the heater fuse **38** to be fused.
(35) The heater fuse **38** is connected between the positive electrode terminal TA, which is one of the pair of output terminals, and the positive electrode terminal corresponding to the positive electrode terminal TA, that is, between the positive electrode terminal TA and a positive electrode terminal T**1** of the battery unit **30** on the same polarity side. The heater fuse **38** is allowed to be fused in accordance with the control of the control IC **35**.
(36) The battery unit **40**, which is an example of the second battery unit, includes a battery block **41** and a battery block **42** connected in series. Specific configuration examples of the battery block **41** and battery block **42** will be described later. In addition, the battery unit **40** has a positive electrode terminal T**3** and a negative electrode terminal T**4**.
(37) The control IC **45** is connected to the battery unit **40** and the heater fuse **48**. The control IC **45** determines the state of each of the battery block **41** and battery block **42**. For example, the control IC **45** measures the voltage for each of the battery block **41** and battery block **42**, and determines whether each battery block is overcharged or not. Then, if there is any battery block overcharged, the control IC **45** controls the current to the heater of the heater fuse connected in series to the battery block, that is, the heater fuse **48**. This control allows the heater fuse **48** to be fused.
(38) The heater fuse **48** is connected between the positive electrode terminal TA, which is one of the pair of output terminals, and the corresponding positive electrode terminal, that is, between the positive electrode terminal TA and a positive electrode terminal T**3** of the battery unit **40** on the same polarity side. The heater fuse **48** is allowed to be fused in accordance with the control of the control IC **35**.
(39) The fuse **50** is another fuse that is different from the heater fuse **38** and the heater fuse **48**. As the fuse **50**, a common fuse such as a current fuse can be applied. The connection midpoints of the battery blocks in each of the different battery units are connected to each other via the fuse **50**. Specifically, the connection midpoint P**1** between the battery block **31** and the battery block **32** in the battery unit **30** and the connection midpoint P**2** between the battery block **41** and the battery block **42** in the battery unit **40** are connected via the fuse **50**. It is to be noted that the connection midpoint in the present specification refers to an arbitrary position in a connection path in a circuit sense, which is not necessarily required to be the center in the connection path.
(40) (In Regard to Heater Fuse)
(41) Next, an example of the heater fuse will be described. As illustrated in FIG. **6**A, the heater fuse **38** includes two fuses **38**A and **38**B connected in series between the positive electrode terminal TA and the positive electrode terminal T**1**, and a heater **38**C that has one end side connected to a connection midpoint P**38** between the fuses **38**A and **38**B. The heater **38**C is a heat generator such as a resistance heater.
(42) The fuse **38**A has one end side (side opposite to the connection midpoint P**38**) connected to the ground via a battery E**1**. In addition, the other end side of the heater **38**C is connected to the ground via the switch SW**1**. The switch SW**1** is a switch that is turned on/off by the control IC **35**, and is provided near the heater fuse **38**. The battery E**1** is, for example, a battery block overcharged (details will be described later).
(43) Normally, the switch SW**1** is turned off. When there is a need to fuse the fuse **38**A and the fuse

38B, the control IC **35** performs control to operate the heater fuse **38**. Specifically, the switch SW**1** is turned on by the control IC **35**. When the switch SW**1** is turned on, a current flows through a current loop RPA in FIG. **6**A. The control performed by the control IC **35** to cause the current to flow causes a current to flow through the heater **38**C, thereby causing the heater **38**C to generate heat. The fuse **38**A and the fuse **38**B are fused by the heat. As described above, the heater fuse **38** is a device that is capable of, as with a normal current fuse or the like, fusing due to an excessive current flowing through the fuse, and also causing a current to flow through the heater **38**C to fuse the fuse **38**A and the fuse **38**B at a predetermined timing.
(44) As illustrated in FIG. **6**B, the heater fuse **48** includes two fuses **48**A and **48**B connected in series between the positive electrode terminal TA and the positive electrode terminal T**3**, and a heater **48**C that has one end side connected to a connection midpoint P**48** between the fuses **48**A and **48**B. The heater **48**C is a heat generator such as a resistance heater.
(45) The fuse **48**A has one end side (side opposite to the connection midpoint P**48**) connected to the ground via a battery E**2**. The other end side of the heater **48**C is connected to the ground via the switch SW**2**. The switch SW**2** is a switch that is turned on/off by the control IC **45**, and is provided near the heater fuse **48**. The battery E**2** is, for example, a battery block overcharged (details will be described later).
(46) Normally, the switch SW**2** is turned off. When there is a need to fuse the fuse **48**A and the fuse **48**B, the control IC **45** performs control to operate the heater fuse **48**. Specifically, the switch SW**2** is turned on by the control IC **45**. When the switch SW**2** is turned on, a current flows through a current loop RPB in FIG. **6**B. The control performed by the control IC **45** to cause the current to flow causes a current to flow through the heater **48**C, thereby causing the heater **48**C to generate heat. The fuse **48**A and the fuse **48**B are fused by the heat. As described above, the heater fuse **48** is a device that is capable of, as with a normal current fuse or the like, fusing due to an excessive current flowing through the fuse, and also causing a current to flow through the heater **48**C to fuse the fuse **48**A and the fuse **48**B at a predetermined timing.
Detailed Configuration Example of Battery Pack According to Present Embodiment
(47) FIG. **7** is a diagram illustrating a detailed configuration example of the battery pack **10** according to the present embodiment. The battery unit **30** includes, for example, fourteen battery blocks (battery block **311***a*, battery block **311***b*, . . . battery block **311***n*) connected in series. It is to be noted that when there is no need to distinguish the individual battery blocks, the battery blocks are appropriately abbreviated as battery blocks **311**.
(48) The battery block **311** has a configuration with four batteries connected in parallel. The number of batteries connected in parallel can be appropriately changed, but is appropriately set to the extent that the temperatures of the batteries are not equal to or higher than a certain level due to a short-circuit current generated when any of the batteries is internally short-circuited.
(49) The battery unit **40** includes, for example, fourteen battery blocks (battery block **411***a*, battery block **411***b*, . . . battery block **411***n*) connected in series. It is to be noted that when there is no need to distinguish the individual battery blocks, the battery blocks are appropriately abbreviated as battery blocks **411**.
(50) The battery block **411** has a configuration with four batteries connected in parallel. The number of batteries connected in parallel can be appropriately changed, but is appropriately set to the extent that the temperatures of the batteries are not equal to or higher than a certain level due to a short-circuit current generated when any of the batteries is internally short-circuited.
(51) The fuse **50** includes (the number of battery blocks—1) fuses, that is, thirteen fuses (fuse **50***a*, fuse **50***b*, . . . fuse **50***m*). For example, the connection midpoint between the battery block **311***a* and the battery block **311***b* and the connection midpoint between the battery block **411***a* and battery block **411***b* of the battery unit **40** are connected via the fuse **50***a*. The connection midpoints of the other battery blocks are similarly connected to each other via the fuses.
(52) In addition, the battery pack **10** includes an AFE (Analog Front End) **71**, an MPU (Micro

Processing Unit) **72**, and a charge/discharge control switch **73**. The charge/discharge control switch **73** includes a charge control switch **73**A and a discharge control switch **73**B.
(53) The AFE **71** is a type of protection IC, which measures the voltage of each battery block, and a charge current and a discharge current flowing through the circuit in the battery pack **10**. Then, the AFE **71** transmits the measurement results to the MPU **72**, which is an upper-level IC.
(54) The MPU **72** comprehensively controls operations such as a protection operation of the battery pack **1**. In addition, the MPU **72** communicates with an electronic device side (host side) to which the battery pack **10** is applied, via a communication terminal COM.
(55) The charge/discharge control switch **73** is a switch whose on/off is controlled by the AFE **71**. As the charge/discharge control switch **73**, for example, an FET (Field Effect Transistor) can be used. The charge/discharge control switch **73** is connected in series to the positive electrode terminal TA and the heater fuse **38** or heater fuse **48** on a power line PLA on the positive electrode side.
Operation Example of Battery Pack
(56) (On/Off Control of Charge/Discharge Control Switch)
(57) Next, an operation example of the battery pack **10** according to the present embodiment will be described. First, on/off control for the charge/discharge control switch **73** will be described.
(58) As described above, the AFE **71** transmits, to the MPU **72**, the voltage of each battery block, and the results of measuring the charge current and discharge current flowing through the circuit in the battery pack **10**. The MPU **72** determines whether it is necessary to block the current path or not, based on the transmitted measurement results. If it is necessary to block the current path, the MPU **72** outputs a control command to the AFE **71**. In response to the control command, the AFE **71** appropriately turns on/off the charge control switch **73**A and the discharge control switch **73**B, thereby blocking the current path.
(59) For example, if the MPU **72** determines that the battery section **20** without any abnormality or the like can be charged and discharged without any problem, the MPU **72** controls the AFE **71** to turn on the charge control switch **73**A and the discharge control switch **73**B. In addition, the MPU**72** controls the AFE **71** to turn off at least the charge control switch **73**A when it is necessary to prohibit charging, for example, when the voltage of the battery section **20** reaches the overcharge prohibition voltage. In addition, the MPU**72** controls the AFE **71** to turn off at least the discharge control switch **73**B when it is necessary to prohibit discharging, for example, when the voltage of the battery section **20** reaches the overdischarge prohibition voltage. In addition, when the battery section **20** is deeply discharged to reach the recharge inhibition region, the MPU **72** controls the AFE **71** to turn off the charge control switch **73**A and the discharge control switch **73**B, thereby stopping the charging/discharging.
(60) (Fusing Control for Heater Fuse)
(61) Next, fusing control for the heater fuse **38** and the heater fuse **48** will be described. It is to be noted that in the following description, the configurations of the battery unit **30** and battery unit **40** illustrated in FIG. **5** will be taken as an example to describe the fusing control for the heater fuses in consideration of convenience of description. In addition, the description will be given on the assumption that the battery block **42** of the battery unit **40** is internally short-circuited.
(62) After the battery block **42** is internally short-circuited, a current from the battery block **32** flows through the fuse **50**, thereby fusing fuse **50**. Then, a current from the battery unit **30** flows to the battery unit **40** via the heater fuse **38** and the heater fuse **48**.
(63) In this regard, when the heater fuse **48** is fused prior to the heater fuse **38**, the battery block **42** internally short-circuited can be electrically separated. Accordingly, if a charge voltage is applied to the battery pack **1** via the positive electrode terminal TA and the negative electrode terminal TB, the safety of the battery pack **1** can be secured. In contrast, when the heater fuse **38** is fused prior to the heater fuse **48**, the battery block **42** fails to be electrically separated. When the battery pack **1** is charged in this condition, the voltage of the battery block **42** is not increased because of the internal

short circuit generated, and only the voltage of the battery block **41** is increased. The increased voltage of the battery block **41** brings the battery block **41** into an overcharged state (state in which the voltage is equal to or higher than the threshold value (for example, 4.3 V in the case of the battery)).

(64) The overcharged state of the battery block **41** is detected by the control IC **45**. The control IC **45** that has detected the overcharged state of the battery block **41** determines that it is necessary to operate the heater fuse **48** connected in series with the battery block **41** in the overcharged state. Then, the control IC **45** performs control to cause a current to flow through the heater fuse **48** (control to turn on the switch SW**2** described above). This control allows a current to flow through the heater **48**C with the use of the electric power of the battery block **41**, thereby allowing the heat of the heater **48**C to fuse the fuse **48**A and the fuse **48**B. Accordingly, the battery unit **40** including the battery block **42** can be electrically separated.

(65) Similar processing is performed also if the battery block other than battery block **42** is internally short-circuited. For example, when the battery block **32** is internally short-circuited, and an overcharged state of the battery block **31** is then detected, the control IC **35** performs control to operate the heater fuse **38**. In addition, similar processing is performed if any of the batteries constituting the battery block **311** and the battery block **411** is internally short-circuited.

Advantageous Effect

(66) According to the present embodiment described above, for example, the following effects can be obtained.

(67) The battery block internally short-circuited can be reliably electrically separated, thus allowing heat generation, ignition, and the like due to the internal short circuit of the battery block to be prevented, and allowing the safety of the battery pack to be secured.

(68) In the specific example described above, when the heater fuse **48** is fused prior to the heater fuse **38**, none of the battery blocks is brought into an overcharged state, and the control is thus not performed to fuse the heater fuse **38**. Accordingly, the use of the battery unit **30** can be continued.

Modification Example

(69) While the embodiments of the present invention have been concretely described above, the contents of the present invention are not to be considered limited to the embodiments described above, and it is possible to make various modifications based on technical idea of the present invention.

(70) In the embodiment described above, the control IC may determine a state other than the overcharged state, thereby determining the necessity of operating the heater fuse. For example, the control IC may determine the state of the voltage balance between the battery blocks. When charging is performed, if the voltage of one of the battery blocks connected in series is increased, whereas the voltage of the other battery block is not substantially changed, the control IC may determine that it is necessary to operate the heater fuse because the other battery block is internally short-circuited. Without charging, when the voltage difference between the battery blocks is equal to or larger than a certain level, it may be determined that it is necessary to operate the heater fuse on the assumption that the battery block is internally short-circuited. This control allows the battery with an abnormality caused to be separated earlier. In addition, the control IC may determine the presence or absence of a battery block that undergoes no substantial change in voltage if charging is performed, and determine that it is necessary to operate the heater fuse if there is such a battery block. With these conditions combined, the control IC may determine whether it is necessary to operate the heater fuse or not.

(71) The example of operating the heater fuse with the use of the electric power of the battery block in the overcharged state is described in the embodiment described above, but the invention is not to be considered limited to the example. For example, with a battery pack provided with a constant current source, the flow of a current from the constant current source to the heater of the heater fuse may cause the heater to generate heat. In addition, some or all of the plurality of heater fuses

included in the battery pack may be connected between the negative electrode terminal of the battery pack and the negative electrode terminal of the battery unit. In addition, in preparation for operational failures of the heater fuses, the heater fuses may be connected both between the positive electrode terminal of the battery pack and the positive electrode terminal of the battery unit and between the negative electrode terminal of the battery pack and the negative electrode terminal of the battery unit.

(72) In the embodiment described above, the AFE may autonomously determine the necessity of the protection operation, based on the measurement result of a voltage or the like, instead of the control by the MPU, and appropriately turn on/off the charge/discharge control switch in response to the determination result. In addition, the MPU may transmit the measurement result transmitted from the AFE, to the electronic device side, thereby causing a microcomputer on the electronic device side to determine the necessity of the protection operation.

(73) In the embodiment described above, the control IC may communicate with the AFE or the MPU. For example, the control IC may communicate with the MPU, thereby causing the control IC to notify the MPU that a predetermined heater fuse has been fused. Then, the MPU may notify the user to prompt the replacement of the heater fuse, or may display or sound an error message or the like.

(74) In the embodiment described above, the charge/discharge control switch may be connected in series with the negative electrode terminal TB and the heater fuse connected to the negative electrode side of the battery unit on the power line PLB on the negative electrode side. In addition, the configuration of the battery pack, such as the configuration of the control IC and AFE integrated, can be appropriately changed without departing from the scope of the present invention. In addition, the number of the battery units connected, the number of the battery blocks connected, the configurations of the battery blocks (the numbers of batteries and the connection modes), and the like in the embodiment described above can be appropriately changed.

(75) The matters described in the above-described embodiments and modification examples can be appropriately combined. In addition, the materials, processes, and the like described in the embodiments are considered merely by way of example, and the contents of the present invention are not to be considered limited to the exemplified materials or the like.

Application Example

(76) The battery packs **10** according to the present invention can be used for mounting on an electric tool, an electric vehicle, various electronic devices, or the like, or for supplying electric power thereto.

(77) FIG. **8** schematically illustrates a configuration example of a hybrid vehicle (HV) that employs a series hybrid system to which the present invention is applied, as an example of applying the present invention to an electric storage system for an electric vehicle. The series hybrid system is intended for a vehicle that runs on an electric power-driving force conversion device, with the use of electric power generated by a generator powered by an engine, or the electric power stored once in the battery.

(78) The hybrid vehicle **600** carries an engine **601**, a generator **602**, the electric power-driving force conversion device (direct-current motor or alternate-current motor, hereinafter referred to simply as a “motor **603**”), a driving wheel **604***a*, a driving wheel **604***b*, a wheel **605***a*, a wheel **605***b*, a battery **608**, a vehicle control device **609**, various sensors **610**, and a charging port **611**. As the battery **608**, the battery pack **10** according to the present invention can be applied.

(79) The motor **603** is operated by the electric power of the battery **608**, and the torque of the motor **603** is transmitted to the driving wheels **604***a* and **604***b*. The torque produced by the engine **601** makes it possible to reserve, in the battery **608**, the electric power generated by the generator **602**. The various sensors **610** control the engine rotation speed via the vehicle control device **609**, and control the position of a throttle valve, not shown.

(80) When the hybrid vehicle **600** is decelerated by a braking mechanism, not shown, the resistance

force during the deceleration is applied as torque to the motor **603**, and the regenerative electric power generated by the torque is reserved in the battery **608**. In addition, the battery **608** is connected to an external power supply through the charging port **611** of the hybrid vehicle **600**, thereby making charge possible. Such an HV vehicle is referred to as a plug-in hybrid vehicle (PHV or PHEV).

(81) It is to be noted that the secondary battery according to the present invention can also be applied to a downsized primary battery, and then used as a power supply for a pneumatic sensor system (TPMS: Tire Pressure Monitoring System) built in the wheels **604** and **605**.

(82) Although the series hybrid vehicle has been described above as an example, the present invention can be also applied to a parallel system in which an engine and a motor are used in combination or a hybrid vehicle in which a series system and a parallel system are combined. Furthermore, the present invention can be also applied to electric vehicles (EVs or BEVs) that run on driving by only a driving motor without using any engine, and fuel cell vehicles (FCVs). In addition, the present invention is also applicable to an electric bicycle.

DESCRIPTION OF REFERENCE SYMBOLS

(83) **10**: Battery pack **30**, **40**: Battery unit **31**, **32**, **41**, **42**: Battery block **35**, **45**: Control IC **38**, **48**: Heater fuse **38**C, **48**C: Heater **73**: Charge/discharge control switch TA: Positive electrode terminal of battery pack TB: Negative electrode terminal of battery pack T**1**, T**3**: Positive electrode terminal of battery unit T**2**, T**4**: Negative electrode terminal of battery unit

Claims

1. A battery pack comprising: a battery section; a pair of output terminals; a control unit; and a plurality of battery units, wherein each of the plurality of battery units includes a plurality of battery blocks connected in series, a positive electrode terminal, and a negative electrode terminal, the battery block includes one battery or a plurality of batteries connected in parallel, the battery section includes two battery units, a fuse is connected between at least one output terminal of the pair of output terminals and a positive electrode terminal or a negative electrode terminal corresponding to the one output terminal, and the fuse is allowed to be fused by the control unit.
2. The battery pack according to claim 1, wherein the fuse is a heater fuse, and the control unit controls a current to the heater to allow the fuse to be fused.
3. The battery pack according to claim 2, wherein the control unit determines a state of each of the plurality of battery blocks, and controls a current to the heater, based on the determined state.
4. The battery pack according to claim 3, wherein the control unit determines whether each of the plurality of battery blocks is in an overcharged state or not, and controls a current to the heater of the fuse connected in series with the battery block in the overcharged state.
5. The battery pack according to claim 1, wherein a first battery unit and a second battery unit are connected in parallel, and the fuse is connected to each of a positive electrode terminal of the first battery unit and a positive electrode terminal of the second battery unit, or to each of a negative electrode terminal of the first battery unit and a negative electrode terminal of the second battery unit.
6. The battery pack according to claim 1, comprising a plurality of the control units, wherein the control units are connected to each of the plurality of battery units.
7. The battery pack according to claim 1, wherein connection midpoints of the battery blocks in each of the different battery units within the battery section are connected to each other via another fuse that is different from the fuse.
8. The battery pack according to claim 1, wherein a charge/discharge control switch is connected in series between one of the pair of output terminals and the fuse.
9. An electric vehicle comprising the battery pack according to claim 1.

10. The battery pack according to claim 1, comprising a plurality of battery sections, each of which includes a first battery unit and a second battery unit.